

Rank stability for normal matrices of bounded spectral complexity

Candidate partial-result packet for arXiv:1708.05338

17 August 2026

Outcome. Fix the tuple length n and a spectral-complexity bound M . Pairwise almost commuting normal matrices, each having at most M distinct eigenvalues, are uniformly close in normalized rank to an exactly commuting tuple. Invertibility is unnecessary. The conclusion persists if all but a sufficiently small rank fraction of each spectrum is supported on at most M values.

1. The source question and the partial answer

For $X, Y \in \text{Mat}_d(\mathbb{C})$ write

$$d_{\text{rk}}(X, Y) = \frac{\text{rank}(X - Y)}{d}, \quad \|[X, Y]\|_{\text{rk}} = \frac{\text{rank}(XY - YX)}{d}.$$

Elek–Grabowski prove stability when every matrix is self-adjoint or unitary, and ask for arbitrary normal invertible matrices.

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Remark 2. It is natural to ask whether the matrices $B_1, \dots, B_n$ can be taken to be “of the same type” as the matrices $A_1, \dots, A_n$, e.g. whether we can demand, say, the matrix B_1 to be unitary, provided that A_1 is unitary. We do not know the answer to this question.

We think that Theorem 1 likely stays true when $A_1, \dots, A_n$ are allowed to be arbitrary normal invertible matrices. On the other hand, it would be interesting to find a counterexample when $A_1, \dots, A_n$ are allowed to be arbitrary invertible matrices.

Figure 1: The exact questions in Remark 2, source PDF page 3.

Theorem 1 (Bounded spectral complexity). *For every $n, M \geq 1$ and $\varepsilon > 0$ there is $\delta > 0$, independent of d , with the following property. If $A_1, \dots, A_n \in \text{Mat}_d(\mathbb{C})$ are normal, $|\text{spec}(A_i)| \leq M$, and*

$$\|[A_i, A_j]\|_{\text{rk}} \leq \delta \quad (1 \leq i, j \leq n),$$

then there are pairwise commuting $B_1, \dots, B_n \in \text{Mat}_d(\mathbb{C})$ such that $d_{\text{rk}}(A_i, B_i) \leq \varepsilon$ for every i .

Proof intuition. On a finite spectrum, complex conjugation is exactly a polynomial: if A is normal with at most M spectral values, then $A^* = p(A)$ for some $\deg p < M$. A telescoping identity bounds the rank cost of passing from A to $p(A)$ solely in terms of M . Thus adjoining all adjoints produces the star-closed almost commuting tuple required by the source’s general theorem.

2. Polynomial commutators

Lemma 1. *Let $p \in \mathbb{C}[z]$ have degree at most m . For square matrices X, Y ,*

$$\text{rank}[p(X), Y] \leq \frac{m(m+1)}{2} \text{rank}[X, Y].$$

Proof. For $\ell \geq 1$,

$$[X^\ell, Y] = \sum_{q=0}^{\ell-1} X^q [X, Y] X^{\ell-1-q},$$

so its rank is at most $\ell \operatorname{rank}[X, Y]$. Expand p and use subadditivity of rank. Scalar coefficients do not affect the estimate. $\square$

Proof of Theorem 1. Put $K = \max\{1, M(M-1)/2\}$. Apply Theorem 6 of Elek–Grabowski to tuples of length $2n$ and tolerance ε , and call its commutator tolerance η ; choose $\delta = \eta/K^2$.

By the spectral theorem, for each i there is a polynomial p_i of degree at most $M-1$ such that

$$p_i(\lambda) = \bar{\lambda} \quad (\lambda \in \operatorname{spec}(A_i)), \quad \text{hence} \quad A_i^* = p_i(A_i).$$

Lemma 1 gives

$$\begin{aligned} \operatorname{rank}[A_i^*, A_j] &\leq K \operatorname{rank}[A_i, A_j], \\ \operatorname{rank}[A_i^*, A_j^*] &\leq K^2 \operatorname{rank}[A_i, A_j]. \end{aligned}$$

The second inequality is obtained by applying the lemma once to each polynomial. Thus $(A_1, \dots, A_n, A_1^*, \dots, A_n^*)$ is a star-closed η -commuting tuple in normalized rank. The source's Theorem 6 supplies a commuting $2n$ -tuple within ε ; its first n entries are the required B_i . No step uses invertibility. $\square$

3. A robust spectral-concentration upgrade

Corollary 1. *For every $n, M \geq 1$ and $\varepsilon > 0$ there exist $\delta, \kappa > 0$, independent of d , such that the following holds. Suppose the normal matrices $A_1, \dots, A_n \in \operatorname{Mat}_d(\mathbb{C})$ satisfy $\|[A_i, A_j]\|_{\operatorname{rk}} \leq \delta$, and for each i some set $S_i \subset \operatorname{spec}(A_i)$ of at most M values has total eigenspace dimension at least $(1 - \kappa)d$. Then the conclusion of Theorem 1 holds.*

Proof. Use Theorem 1 with tolerance $\varepsilon/2$, and denote its input tolerance by η . Choose

$$\delta \leq \eta/2, \quad \kappa \leq \min\{1/2, \varepsilon/2, \eta/8\}.$$

In an orthonormal eigenbasis for A_i , replace every eigenvalue outside S_i by one fixed member of S_i ; call the resulting normal matrix C_i . Then $|\operatorname{spec}(C_i)| \leq M$ and $d_{\operatorname{rk}}(A_i, C_i) \leq \kappa$. If $D_i = C_i - A_i$, then

$$[C_i, C_j] - [A_i, A_j] = [D_i, C_j] + [A_i, D_j],$$

and consequently

$$\|[C_i, C_j]\|_{\operatorname{rk}} \leq \delta + 2d_{\operatorname{rk}}(A_i, C_i) + 2d_{\operatorname{rk}}(A_j, C_j) \leq \eta.$$

Apply Theorem 1 to the C_i and use the rank triangle inequality. $\square$

4. Why this does not yet prove the full conjecture

The dependence on M is genuine in this argument: interpolating the adjoint on an unrestricted finite spectrum can require degree $d-1$, destroying the dimension-free commutator modulus. Nor can the source's one-line unitary/self-adjoint reduction simply be copied. Indeed, let

$$v = (1, 2, 3, 4)^T, \quad U = I - \frac{2vv^*}{v^*v},$$

$$A = \operatorname{diag}(0, 1, -5i, 2+i), \quad B = U \operatorname{diag}(0, 1+i, 2-i, 3+2i)U^*.$$

Both A and B are normal, but exact row reduction gives

$$\operatorname{rank}[A, B] = 2, \quad \operatorname{rank}[A^*, B] = 4.$$

Thus even equality of the two commutator ranks fails for normal pairs. This example obstructs the simplest rank-Fuglede shortcut; it is not a counterexample to stability.

The theorem also does not answer the separate same-type question: Theorem 6 produces arbitrary commuting approximants, with no guarantee that they remain normal or invertible.

Upgrade audit. Eight focused stages were completed: direct star-closure; later-literature audit; exact rank-Fuglede stress test; spectral interpolation; quantitative Horner control; star-closed completion; spectral-concentration upgrade; and unrestricted/same-type obstruction analysis. The successful route closes every uniformly bounded spectral-complexity class. The remaining barrier is dimension-free control of adjoints for spectra whose cardinality grows with d .

Novelty boundary. A 2024 primary source still records the unrestricted spectral-restriction-free problem as open. Bounded searches through 17 August 2026 found no paper stating this partial theorem. Expert review and a broader priority search are required before dissemination.

References

- [1] G. Elek and L. Grabowski, *Almost commuting matrices with respect to the rank metric*, arXiv:1708.05338 (2017; v3, 2021).
- [2] T. Bauer, G. Blachar, and B. Greenfeld, *Rank-stability of polynomial equations*, arXiv:2401.04676 (2024).